



Predictors of teachers' intention to implement inclusive education

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ABSTRACT

Teachers are among the most important players in the successful implementation of inclusive education. This study aimed to examine Saudi Arabian teachers' intentions to implement inclusive education and the influence of their demographics, attitudes, self-efficacy, perception of support, and concerns related to inclusive education on their intention. This study included 125 in-service special and general education teachers. The regression analysis results revealed that teachers' self-efficacy and majors (special education vs. general education) significantly influenced their intentions to implement inclusive education in regular classrooms. Finally, we discuss the results and their implications for research and practice.

1. Introduction

Internationally, the United Nations Educational, Scientific, and Cultural Organisation (Unesco, 2000) supports inclusive education for all students; they should be able to learn together in a safe, supportive, accepted, encouraged, and quality-inclusive learning environment (Bui et al., 2010). Inclusive education focuses on removing barriers that may prevent students from reaching their full potential, allowing them to access the same learning opportunities and helping reduce the achievement gap between different student groups (Ainscow & Miles, 2008). Much evidence in the literature confirms that inclusive education benefits all students, including those with disabilities (Ainscow, 2007; Al-Musa, 2008; Szumski et al., 2022). Inclusive education helps create a more positive school climate and a culture of acceptance, improves academic achievement, and fosters social skills and relationships among students (Alquraini & Gut, 2012; Bui et al., 2010; Cole et al., 2004; Dupuis et al., 2006; Newman, 2006; Szumski et al., 2022).

Teachers play an important role in successfully implementing inclusive education principles and practices in classrooms (Carew et al., 2019; Forlin & Lian, 2008; Sharma et al., 2021). The importance of teachers' roles in inclusive education has prompted researchers to investigate variables that may impact teachers' practices related to inclusive education. Studies have analysed and examined multifaceted environmental and personal factors that may impact inclusive education and teachers' practices. Teachers' intentions, attitudes, support, self-efficacy, and concerns affect their inclusive educational practices (Hellmich et al., 2019; Opoku et al., 2021; Savolainen et al., 2012; Sharma et al., 2021; Wray et al., 2022). We discuss some of these factors

and provide an overview of inclusive education in Saudi Arabia.

1.1. Teacher intention

Teachers' intentions towards inclusive education are prerequisites for implementing inclusive education practices (Hellmich et al., 2019; Opoku et al., 2021; Yan & Sin, 2014), and motivating actual behaviour (Ajzen, 2011). Researchers have found that several factors, such as teachers' attitudes (Yan & Sin, 2015), school environment, self-efficacy (Hellmich et al., 2019; Opoku et al., 2021), provision of training and external support, a shared vision among teachers, and global and national policies related to the rights of individuals with disabilities (Mokaleng & Möwes, 2020) positively influence teachers' intentions, which in turn affect the implementation of inclusive education practices (Opoku et al., 2021; Wray et al., 2022). However, the influence of these factors may vary in different cultural contexts (Forlin, 2007; Opoku et al., 2021).

1.2. Teacher's self-efficacy for inclusive education

Teachers' confidence in their capacity to teach and encourage student learning is known as teacher self-efficacy. A key element of inclusive education implementation is teachers' self-efficacy related to inclusive education (Kristiana, 2018; Opoku et al., 2021; Woodcock & Jones, 2020). Teachers with higher levels of self-efficacy are more likely to successfully implement inclusive educational practices (Capp, 2018; Woodcock et al., 2022; Yan & Sin, 2014). They are also more likely to have higher expectations of students, to be successful in creating and

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maintaining an inclusive learning environment (Buric & Kim, 2020), and to be successful in helping students enhance their achievement and reach their potential (Kiel et al., 2020; Zee & Koomen, 2016). Teachers form the backbone of inclusive education; thus, they must have inclusive education knowledge, skills, resources, and awareness of laws and regulations to ensure that all students have access to the curriculum and are supported in their learning (Wray et al., 2022; Woodcock & Jones, 2020).

Several factors contribute to teachers' self-efficacy in inclusive classrooms. Teachers' self-efficacy for inclusive education is affected by the type of preparation (Chiaro, 2014; Wray et al., 2022), knowledge about inclusion laws and policies (Jung, 2007), previous interactions with people with disabilities, confidence levels in teaching students with disabilities (Loreman, 2013), personality, and school climate, including support from school leaders and colleagues (Kristiana, 2018). Research has shown that teachers' demographics affect their self-efficacy concerning inclusive education (Wray et al., 2022). Some studies have found that male teachers have higher self-efficacy than female teachers in managing behaviour in the classroom (Peebles & Mendaglio, 2014; Specht et al., 2016). Elementary teachers tend to have higher self-efficacy in inclusive education than secondary school teachers (Specht et al., 2016). Those with prior teaching experience and training in working with students with disabilities show higher self-efficacy than those without such experience (Avramidis & Kalyva, 2007; Peebles & Mendaglio, 2014).

1.3. Attitude towards inclusive education

Teachers' attitudes towards inclusive education can affect the implementation of inclusive education (Forlin and Chambers, 2011; Sharma & Jacobs, 2016). Several variables are related to teachers' attitudes towards inclusive education, such as students' disability severity and type (Ahmmed et al., 2012; Savage & Erten, 2015; Sawyer et al., 2022; Wray et al., 2022), training and school support (Alshahrani, & Abu-Alghayth, 2023; Chiner & Cardona, 2013; Werner et al., 2021), experience with inclusive education, knowledge of inclusion policies, and self-efficacy regarding inclusive education (Werner et al., 2021; Woodcock & Jones, 2020). Additionally, Alanazi (2022) found that teachers with master's and Ph.D. degrees held more positive attitudes towards inclusive education than teachers with bachelor's degrees. Furthermore, the cultural context may impact teachers' attitudes towards inclusive education. For example, Alnahdi et al. (2019) compared Saudi Arabian and Finnish pre-service teachers regarding their attitudes towards inclusive education. They found that Saudi pre-service teachers tended to have less positive attitudes toward inclusive education than Finnish teachers because Saudi teachers are more likely to feel the weight of individual responsibility as there is not a strong culture of teamwork in the provision of support for children with disabilities in the Saudi context (Alnahdi et al., 2019). Furthermore, Alnahdi et al. (2019) reported that Saudi and Finnish teachers valued the concept of inclusion similarly.

1.4. Teacher's perception of support

The successful implementation of inclusive education requires the provision of adequate support and resources for teachers to ensure that all students can access the curriculum and participate in the learning process along with each other (Yan & Sin, 2014). Teachers with a high perception of available support tend to be confident in their ability to succeed in meeting students' diverse needs and dealing with challenging situations because they feel that resources and staff support are available when they need assistance (Monsen et al., 2014; Werner et al., 2021). According to the theory of planned behaviour (Ajzen, 2011), a person's perceived support from significant others regarding expectations and support (subjective norms) affects their behaviour. Thus, teachers' perceptions of support may influence their intentions to implement

inclusive educational practices (Ahmmed et al., 2014; Civitillo et al., 2016). Support from schools and others not only impacts teachers' intentions and inclusive education practices but is also associated with teachers' attitudes towards inclusive education (Chiner & Cardona, 2013; Desombre et al., 2021; Horne & Timmons, 2009; Monsen et al., 2014; Werner et al., 2021).

1.5. Concern about inclusive education

Inclusive education is a major challenge facing countries worldwide (Desombre et al., 2021). While inclusive education has many benefits, there are concerns regarding educational systems and schools. One of the main concerns is that accommodating the needs of all students in a single classroom can be difficult. Some critics argue that it can be difficult to ensure that all students receive the same quality of education (Alhammad, 2017). Others claim that inclusive education can decrease the quality of education for students who are more capable (Waitoller & Artiles, 2013). It can be difficult to ensure that all students are treated equally, regardless of their individual differences (Alhammad, 2017). In addition, there may be a lack of understanding and acceptance of inclusive education among students, parents, and other community members, which requires cultural changes to successfully implement inclusive education (Desombre et al., 2021). Some schools face challenges in accommodating students with disabilities, offering support to teachers (Kristiana, 2018; Monsen et al., 2014), having well-trained and qualified school staff (Ainscow, 2005; Chiner & Cardona, 2013; Werner et al., 2021), turning policies on inclusive education into practice (Haug, 2010), and managing inappropriate behaviours (Roberts & Simpson, 2016). These barriers and concerns may impact teachers' intentions to teach in inclusive classrooms (Ahmmed, 2013; Hellmich et al., 2019), negatively affect their attitudes and perceptions about inclusive education (Jury et al., 2023; Song et al., 2019), and restrict the successful implementation of inclusive education (Sharma & Sokal, 2016).

1.6. Inclusive education in Saudi Arabia

Saudi Arabia began including students with disabilities in mainstream schools in 1996 (Al-Mousa, 2010). The process began by including students from special education schools, such as students with intellectual disabilities, autism spectrum disorder, deafness, blindness, and severe disabilities (Al-Mousa, 2010). The implementation of inclusion depends on the type of disability rather than on the needs. Students with learning, physical, behavioural, and emotional disabilities learn in general classrooms, where they participate in the general education curriculum and receive support from special education teachers. Special education support can differentiate instruction by providing viable alternatives to instructional materials, methods, assessments, and learning environments (Almalky & Alwahbi, 2023). Students with intellectual disabilities and autism spectrum disorders still receive services in self-contained classrooms in inclusive schools and participate in general school extracurricular activities (Alquraini, 2012). Both educational settings are available for students with other disabilities, such as hearing and visual impairments, based on the disability severity. In recent years, new initiatives for inclusive education in regular classrooms have been addressed in the strategic plans of the Ministry of Education (Ministry of Education, 2022), focusing on including all students with disabilities in regular classrooms and preparing teachers for inclusive education.

In an effort toward inclusive education in Saudi Arabia, researchers have responded by investigating factors related to inclusive education. Several studies have examined the barriers to inclusive education, whereas others have examined teachers' competence, attitudes, and efficacy related to inclusive education. Some barriers to implementing inclusive education practices include a lack of specialised teacher preparation and training, lack of competence among teachers, lack of collaboration, negative views among some teachers, lack of resources, and crowded classrooms (Alanazi, 2012; Alhammad, 2017; Alharbi &

Madhesh, 2018; Sakz & Woods, 2015). Saudi teachers exhibit high levels of trust in their capacity to work in inclusive environments (Alquraini, 2012; Yada & Alnahdi, 2021). They also expressed concerns about some necessary skills, such as integrating families in school events and handling students with physically aggressive behaviours (Alnahdi, 2020). Other studies have found that Saudi teachers hold neutral to negative attitudes towards inclusive education (Alnahdi et al., 2019). Alqahtani and Rababah (2019) found that teachers believed they were prepared and competent in inclusive education but that the school environment and resources still needed improvement.

Although inclusive education in regular classrooms is still in the early stages of development in Saudi Arabia, it is vital to understand teachers' intentions to implement inclusive education practices and the factors that may predict them, especially because there have been no studies to fill this gap in Saudi Arabia. Thus, this study aimed to examine the influence of teachers' attitudes towards inclusive education, self-efficacy, concerns, and a perception of support regarding inclusive education on their intention to implement inclusive education practices in Saudi Arabia. This study aimed to answer the following questions:

- (1) Do years of experience, gender, specialisation, and school level influence teacher's intention towards practising inclusive education?
- (2) Do teachers' attitudes towards inclusive education, self-efficacy for inclusive education, sense of support for inclusive education, and concern towards inclusive education predict teachers' intention towards practising inclusive education, controlling for demographic variables (experience, specialisation, and school level)?

2. Methods

This study used a cross-sectional survey design to answer the research questions.

2.1. Participants

After obtaining approval for the study, we invited special and general education teachers in Riyadh City to participate in an online survey at their schools. In total, 125 teachers voluntarily participated in this study. Approximately 56 % of the participating teachers were general education, and 44 % were special education teachers. The majority (63 %) of teachers were male ($n = 79$). Half of the teachers taught in elementary school, 28 % in high school, 16 % in middle school, and 6 % in preschool. Of these teachers, 74 % held a bachelor's degree, 25 % held a master's degree, and one had a PhD. Teaching experience ranged from one to 32 years, with a mean of 14 years and a standard deviation of 8. The participants participated in an average of five training workshops on inclusive education, with a standard deviation of 10 (minimum = 0, maximum = 50 workshops).

2.2. Instrument

An online self-report survey was used to collect teachers' data on inclusive education. The survey used five scales. Two scales (i.e. teachers' attitudes towards the inclusive education scale –Arabic [TAIS-A] – and teacher efficacy for inclusive practices [TEIP]) were previously adapted from other contexts and validated in Saudi Arabia (Alnahdi, 2019a, 2019b; Alnahdi et al., 2019). The third scale was (intention to teach in inclusive classrooms [ITICS]), developed by Sharma & Jacobs, 2016. The fourth scale (Teachers' Concerns About the Inclusive Education Scale [CIES]) was designed by Sharma & Desai, (2002). The fifth scale was developed to measure teachers' perceptions of support for inclusive education (TPS).

The scales were subjected to several procedures to ensure face, content, and structural validity, except for the first two scales, the TAIS-

A and the TEIP, which have already been validated in several studies in the Saudi context. For the ITICS and CIES, as they were originally written in English, we asked several university professors with expertise in inclusive education and English speakers to review them and the translated versions to ensure the authors' translation. We then asked them to provide feedback on the ITICS, CIES, and TPS. After modification based on their feedback, several teachers reviewed those scales for readability and clarity. For factorial validity, several explanatory factor analyses (EFAs) were conducted using the maximum likelihood estimation with promax rotation to examine the factorial structure of each of the three scales. For factor extraction, we relied on keeping any factors with an eigenvalue of 1.0 or higher, examining the scree plot to look for a change in the slope, and suggesting a parallel analysis. The final criterion relied on the maximum interpretability of the number of factors in which (a) at least two items had significant loading on one factor (the cut-off point was > 0.35), (b) the items shared conceptual meaning, (c) items loaded on different factors to indicate different constructs, and (d) the item loadings demonstrated a simple structure.

2.2.1. Demographic variable

The demographic section collects information about teachers' gender, major (special education or general education), years of experience in teaching, school level, number of professional development courses, and highest educational attainment.

2.2.2. Teacher intention

Teachers' intentions to teach in an inclusive education classroom were measured using an adapted version of the ITICS developed by Sharma and Jacobs (2016). The original scale had seven items, with response options measured on a seven-point Likert scale (1 = *extremely unlikely* to 7 = *very likely*). Example items included "change the curriculum to meet the learning needs of a student with learning difficulty enrolled in your class" and "consult with the parents of a student who is struggling in your class". Sharma and Jacobs (2016) reported that the scale had two factors, representing a change in curriculum and the intention to consult, with a reliability of approximately 0.7. In this study, the experts and teachers suggested slight modifications to certain wording items. Parallel analysis, scree plots, and Kaiser eigenvalues greater than one criterion suggested one- and two-factor solutions. A two-factor solution satisfied the maximum interpretability criteria. Item "change the assessment tasks to suit the learning profile of a student who is struggling" was cross-loaded and was deleted. The two-factor solution achieved the maximum interpretability and explained 58 % of the variance. Factor one had three items and explained 33 % of the variance. This factor represents the intention to consult with others and improve oneself. Factor two had three items and explained 25 % of the variance. This factor represented the intention to change the instruction and curriculum and involved students with disabilities in the class. The factor correlation was ($r = 0.77$). In this study, Cronbach's alpha for this scale was 0.84.

2.2.3. Teacher's perception of support

TPS scale asked teachers about their perceptions of the support for inclusive education if it was implemented at their school using a 12-item scale. This scale was specifically developed for this study. The authors constructed a draft of 14 items, with response options measured on a five-point Likert (1 = *strongly disagree* to 5 = *strongly agree*). One item was deleted based on feedback. Parallel analysis suggested three factors, whereas a scree plot and Kaiser's eigenvalue greater than one criterion suggested a one-factor solution. One-factor, two-factor, and three-factor solutions were examined. A three-factor solution appears to be better at achieving maximum interpretability. Item "*the availability of curriculum that can be modified to meet all students*" cross-loaded on two factors and deleted. The analysis was repeated. The first factor had six items representing support from the school administration and policy. Example items include "*policy support the implementation of inclusive education*"

and “supervisors provide adequate support to implement best practices to teach all students”. Factor two had three items representing support related to positive attitudes towards inclusive education. Example items included “school personal hold supportive attitude towards inclusive education”. Factor three included three items representing the perception of support related to natural support from colleagues and the school administration. Example items include “teachers support and help each other when needed”. These three factors explained 67 % of the variance, with factor one explained 29 %, factor two 22 %, and factor three 16 %. Factor one was correlated with factor two ($r = 0.75$) and with factor three ($r = 0.66$), while the correlation between factor two and three was ($r = 0.68$). In this study, Cronbach’s alpha for the support scale was 0.94.

2.2.4. Concern about inclusive education

Teachers’ concerns about inclusive education were measured using an adapted version of CIES (Sharma & Desai, 2002). The original scale had 21 items, with response options measured on a four-point Likert scale (1 = *not concerned at all*, 2 = *a little concerned*, 3 = *very concerned*, and 4 = *extremely concerned*). The 21 items on the scale fall under four factors: 1) *concern about resource* (six items), 2) *concern about acceptance* (five items), 3) *concerned about academic standards* (five items), and 4) *concerned about workload* (five items). Sharma and Desai (2002) reported a Cronbach’s alpha of 0.91 for the total scale.

After the process content and face validities, we modified several items according to the suggestions of university professors and teachers. We examined the factorial structure using EFA. Parallel analysis suggested a four-factor solution, while a scree plot and a Kaiser eigenvalue greater than one criterion suggested a three-factor solution. Three-factor, four-factor, and five-factor solutions were examined. However, the four-factor solution seemed to explain the factorial structure better than the three- and five-factor solutions. After promax rotation of the four-factor solution, seven items were cross-loaded. These items were excluded from further analysis. Those items included “I will not receive enough incentives...”, “my school will not have enough fund ...”, “my workload will increase”, “other staff member of the school will be stressed”, “the overall academic standards of the school will suffer”, “my performance as a classroom teacher will decline”, and “difficulty coping with children with disabilities who don’t have adequate self-care skills”. We repeat the analysis using the final 14 items. The Four-factor solution achieved the maximum interpretability and explained 62 % of the variance. Factor one had five items representing teachers’ concerns about the lack of resources. The second factor had three items related to concerns about academics and attention for all students. Factor three included two items related to concerns about acceptance. The fourth factor had four items relating to the teacher’s personal workload and knowledge. These factors have conceptual meanings similar to those of the original scale. We allowed the factors to correlate. Factor one was correlated with factor two ($r = 0.59$), with factor three ($r = 0.55$), and with factor four ($r = 0.59$). Factor two was correlated with factor three ($r = 0.56$), and with factor four ($r = 0.50$). Factor three was correlated with factor four ($r = 0.52$). The overall Cronbach’s alpha for the concern scale was 0.90.

2.2.5. Teacher’s self-efficacy for inclusive education

Teachers’ efficacy in working in an inclusive education setting was measured using the Arabic version of the TEIP (Alnahdi, 2019a). The original scale (Sharma et al., 2012) has been used in several cultural contexts (Ahsan et al., 2012; Park et al., 2016; Yada & Savolainen, 2017) and validated in the Saudi Arabia context (Alnahdi 2019a, 2019b; 2020). The original scale has 18 items, with response options measured on a six-point Likert (1 = *strongly disagree*, to 6 (*strongly agree*). Alnahdi (2019a, 2019b) utilised several procedures to adapt and validate an Arabic version of the scale in the context of Saudi Arabia, including back-translation, pilot testing, Rasch analysis, EFA, and confirmatory factor analysis. The Arabic versions (full and shortened versions; Alnahdi, 2019a, 2019b) showed good psychometric properties and a factorial structure similar to the original scale. We decided to use the

short version of the scale with 13-items in this study because it showed better psychometric properties than the 18- items version in terms of dimensionality (Alnahdi, 2019a). The short version had internal reliability using a person separation index of 0.88. In this sample, Cronbach’s alpha for the short version of the Arabic scale was 0.92.

2.2.6. Attitude towards inclusive education

Teachers’ attitude towards inclusive education was measured using the TAIS-A (Alnahdi et al., 2019). The original scale (Saloviita, 2015) had ten items, with response options measured on a five-point Likert (1 = *strongly disagree*, to 5 (*strongly agree*). After translation to Arabic and pilot testing, Alnahdi et al. (2019) reported acceptable construct and convergent validity of the translated version. Cronbach alpha was 0.69. In this sample, the scale’s reliability using Cronbach’s alpha was 0.77.

2.3. Data analysis

Descriptive and inferential statistics were used to answer the research question. First, descriptive statistics, such as frequency, means, and standard deviations, were calculated for all variables. Second, block entry regression analysis (BRA; Cohen et al., 2013) was utilised to examine the influence of teachers’ attitudes, self-efficacy, concern, and support regarding inclusive education on their intention to implement inclusive education practices above and beyond the variance explained by teachers’ demographics. The order of entry in the regression model was teachers’ years of experience, gender, specialisation or major, and school level in the first block, followed by teachers’ attitude, self-efficacy, concern, and support in the second block. The incremental change in R^2 between the first and second blocks was used to determine the unique contributions of the set of variables in the second block above and beyond the first block line. Prior to the analysis, we examined the linear regression assumptions. Three cases were deemed outliers, with studentised residuals larger than 3. After excluding these cases, 122 teachers were included in the analyses. The assumptions of the regression analysis were met, except for homoscedasticity. We utilised a heteroscedasticity-consistent covariance estimate (HCC3; Zeileis, 2004) to produce robust standard errors for the t-tests of the coefficients and the model F test. All statistical analyses were conducted using R software.

3. Results

The results showed that the teachers in this sample had a relatively high intention to teach in an inclusive education classroom ($M = 6.2$, $SD = 0.67$). Descriptive statistics and Pearson correlations between the quantitative variables are presented in Table 1.

The results of the BRA design on the influence of teachers’ attitudes, self-efficacy, concern, and support regarding inclusive education on their intentions to implement inclusive educational practices are presented in Table 2. First, teachers’ demographics were dummy coded (i. e., gender, school level, and specialisation) and entered into the first regression block. The set of variables in this block was significant and explained 16 % of the variance in teachers’ intentions to implement inclusive educational practices ($R^2 = 0.16$, $F(6,115) = 3.3$, $p < .001$). The partial regression coefficient of the school level for teachers who taught in kindergarten compared to teachers who taught in elementary school (i.e. the reference group) was significant ($B = 0.74$, $t = 3.2$, $p < .001$), suggesting that kindergarten teachers had a higher intention than elementary teachers by 0.74 while holding other variables constant. The partial regression coefficients of school level for teachers who taught at middle and high school compared to teachers who taught at elementary school were not statistically significant while holding other variables constant ($B = -0.27$, $t = -1.4$, $p = .14$; $B = 0.05$, $t = 0.3$, $p = .7$). Teachers’ gender was not statistically significantly associated with teachers’ intention to teach in an inclusive classroom while holding other variables constant (B for female = -0.07 , $t = -0.6$, $p = .54$). The

Table 1
Descriptive and Pearson correlations between the quantitative variables.

Variables	M	SD	1	2	3	4	5
1. Intention	6.2	0.67	–				
2. Self-efficacy	5.0	0.68	0.53**	–			
3. Attitude	2.3	0.64	0.15	0.11	–		
4. Support	3.6	0.89	0.17*	0.21*	0.01	–	
5. Concern	2.3	0.62	-0.24**	-0.31**	-0.24**	-0.50**	–
6. Experience	14.2	8.16	0.08	0.11	0.05	0.23*	-0.18*

Note. * $p < .05$; ** $p < .001$.

Table 2
Block-entry regression analysis with intention as the criterion.

Predictor	Block One			Block Two		
	b	SE b	beta	b	SE b	beta
(Intercept)	5.7			3.5		-0.3
Gender ^{a c}	0.07	0.14	-0.1	0.01	0.13	0.01
Kindergarten (vs elementary) ^c	0.74**	0.23	1.1	0.67*	0.22	1.0
Middle school (vs elementary) ^c	-0.27	0.19	-0.4	-0.20	0.14	-0.31
High school (vs elementary) ^c	0.05	0.15	0.08	0.02	0.14	0.03
Major (special educator) ^{b c}	0.43**	0.12	0.65	0.40**	0.12	0.50
Experience	0.02*	0.01	0.02	0.01	0.01	0.13
Attitude				-0.02	0.09	-0.02
Self-efficacy				0.44**	0.07	0.44
Concern				-0.07	0.10	-0.06
Support				0.08	0.08	-0.10
R ²	0.16			0.40		
F	3.3** (6,115)			9.5** (10,111)		
ΔR ²				0.24		
ΔF				10 (4,111)		

Note. * indicate $p < .05$; ** indicate $p < .01$.

^a 0 = male, 1 = female. ^b 0 = general education, 1 = special education. ^c beta not standardized.

partial regression coefficient for years of experience was significant ($B = 0.02$, $t = 1.99$, $p < .05$), suggesting that, on average, an increase of one year of experience corresponds to a 0.02-point increase in intention while holding other variables constant. Finally, the partial regression coefficient for specialisation was significant (B special educator = 0.43, $t = 3.5$, $p < .001$), suggesting that being a special education teacher corresponded to a 0.43-point increase in intention compared to general education teachers while holding other variables constant.

In the second block, teachers' attitudes, self-efficacy, concerns, and support regarding inclusive education were added to the demographic variables in the first block. The set of demographic variables, teachers' attitudes, self-efficacy, concern, and support, accounted for 40 % of the variance in teachers' intention to implement inclusive education in regular classrooms ($R^2 = 0.40$, $F(10,111) = 9.5$, $p < .001$). Teachers' attitudes, self-efficacy, concern, and support for inclusive education resulted in a significant improvement in explaining the variance in teachers' intentions above and beyond demographic variables ($\Delta R^2 = 0.24$, $\Delta F(4,111) = 10$, $p < .001$). Examination of the partial regression coefficients indicated that only the coefficients for teaching in kindergarten, specialisation, and self-efficacy were significantly associated with teachers' intention to engage in inclusive education while holding the other variables constant. The partial regression coefficient for teaching in kindergarten was still significant but decreased in magnitude compared to the previous model ($B = 0.67$, $t = 2.7$, $p < .05$), suggesting that, on average, being a kindergarten teacher corresponded to a 0.67-point increase in teachers' intention to teach in an inclusive classroom compared to elementary education teachers, while holding other variables constant in the model. The regression coefficient for special education teachers compared to general education teachers decreased but was still significant ($B = 0.40$, $t = 3.6$, $p < .001$), suggesting that special

education teachers corresponded to a 0.40-point increase in their intention to teach in an inclusive classroom compared to general education teachers while holding other variables constant in the model. Finally, the partial regression coefficient for self-efficacy was significant ($B = 0.44$, $t = 5.6$, $p < .001$), indicating that on average, a one-point change in self-efficacy corresponded to an increase in teachers' intention of 0.44-point, while holding other variables constant in the model. None of the other variables were statistically significant.

4. Discussion

In this study, we examined the relationships between teachers' demographic variables (i.e., gender, school level, major, and experience) and their intention to implement inclusive education. We also examined the influence of teachers' attitudes, self-efficacy, perception of support, and concerns related to inclusive education on their intentions to implement inclusive education practices above and beyond demographic variables. Literature indicates that teachers' intentions are a prerequisite for implementing inclusive education practices (Ahmmed et al., 2014; Hellmich et al., 2019; Opoku et al., 2021; Wray et al., 2022). Teachers' intentions to implement inclusive education are influenced by many factors, such as student and teacher demographics, teachers' attitudes, school environment, support, and self-efficacy, which in turn may affect the successful implementation of inclusive education (Sharma & Sokal, 2016; Wray et al., 2022). Several studies have examined teachers' internal attributes, such as self-efficacy (Alquraini, 2012; Alnahdi, 2019a), attitudes (Alnahdi et al., 2019), barriers, and concerns (Alanazi, 2012; Alhammad, 2017; Alnahdi, 2020). However, there is a gap in our understanding of teachers' intentions to implement inclusive education. This study is the first to examine the variables predicting teachers' intentions to implement inclusive education in Saudi Arabia. Investigating these variables added to our understanding of the factors that impact teachers' intention to implement inclusive education in Saudi Arabia, as well as other countries that plan to implement inclusive education.

The results revealed that Kindergarten teachers and teachers majoring in special education were significantly more likely to implement inclusive education, even after accounting for teacher's self-efficacy, attitudes, concern, and perception of support in the second regression block. An interesting finding was that teachers in kindergartens seemed more willing to include students with disabilities in their classrooms than teachers in elementary schools. It could be that kindergarten's teachers have more flexibility, focus less on academic subjects, and are more prepared to work with students with diverse school readiness and abilities (Stemberger and Kiswarday, 2018), which makes them more open to include students with disabilities. Teachers with special education degrees were more inclined to implement inclusive education in regular classrooms than those with other educational degrees. Special education teachers may have better training and competencies (Aldosari, 2022; Alshalhoub, 2019; Miesera et al., 2019), knowledge of inclusion and its policies (Alshahrani, & Abu-Alghayth, 2023; Werner et al., 2021), and positive attitudes towards inclusion than general education teachers, which may transfer to their intentions (Alqahtani & Rababah, 2019; Hellmich et al., 2019; Subban & Mahlo, 2017). Teachers with more experience seemed more willing to

implement inclusive education in regular classrooms, albeit with small effects and lost its significant in the second regression block. It is unclear why this finding, but one explanation could be that teachers with more experience may have more control over their teaching environment and deal with larger numbers of students over the years, which may reduce their concern and increase their attitude (Unianu, 2012). A similar result was found in other studies, in which teachers with more experience had higher intentions and positive attitudes than those with less experience (Ahmmed, 2013; Avramidis & Kalyva, 2007). The results indicated that male and female teachers did not differ in their intentions to implement inclusive education in regular classrooms. This result could be attributed to the environment and teacher preparation in Saudi Arabia, where the male and female have similar educational experiences. Similarly, other studies in Saudi Arabia (Alqahtani & Rababah, 2019) and other societies (Ahmmed, 2013) found that teachers' gender was not associated with their intentions. The influence of teachers' specialisation (special vs. general education teachers) and school level (kindergarten vs. elementary school) shows the importance of examining contextual variables such as school and classroom environment, type of teacher preparation, and interaction with teachers' and students' variables on teachers' intention to implement inclusive education. Teacher qualifications and preparation tend to play major roles in supporting teachers' practices related to inclusive education (Wray et al., 2022; Woodcock et al., 2022). It has become necessary for pre- and in-service educational programs to develop teachers' awareness, attitudes, and competence in inclusive education, regardless of their specialisation.

We hypothesised that constructs drawn from the TPB and others – teachers' self-efficacy, attitudes towards inclusive education, perception of support, and concern – would predict their intentions to implement inclusive education. Although the purpose of the study was not to evaluate the applicability of the TPB in the Saudi context, the results provided a different view of factors predicting the intention to engage in inclusive education that warrant consideration for context in the beginning process of implementing inclusive education.

We found that only teachers' self-efficacy was a significant predictor of their intentions, which aligns with the findings of other studies (Ahmmed et al., 2014; Hellmich et al., 2019; Sharma & Jacobs, 2016; Yan & Sin, 2014). Indeed, a person's confidence in his or her ability plays a major role in the motivation to engage in a specific task or behaviour, as well as in its or actual performance (Ajzen, 1991). Improving self-efficacy is not only a target by itself but also a target for improving intention and actual practices and behaviours (Ajzen, 2011; Sharma et al., 2021; Hellmich et al., 2019). Thus, teachers' self-efficacy related to inclusive education in terms of pedagogical skills, knowledge about inclusive education, interacting with students with disabilities, and finding support and resources must be emphasised in pre- and in-service preparation programs and training, specifically for general education teachers.

Surprisingly, we found that teachers' attitudes, concerns, and support for inclusive education were not significant predictors of intention. These results are inconsistent with other studies conducted using theory such as TPB and others in different cultural contexts. Ahmmed (2013) and Yan and Sin (2014) found that teachers' attitudes were associated with their intentions toward inclusive education. Teachers' attitudes may differ between cultural contexts, and their influence on intentions and practices may also differ (Alnahdi et al., 2019; Forlin, 2007; Opoku et al., 2021). In this study, teachers tended to lean towards negative attitudes, consistent with other studies in Saudi Arabia (Alnahdi et al., 2019; Hakeem, 2009). Thus, it could be that neutral or negative attitudes relate directly to practices rather than intentions to implement inclusive education, as this sample had a high intention for inclusive education. In general, negative attitudes towards disabilities and inclusive education may lead teachers to resist the implementation of inclusive education practices (Hellmich et al., 2019; Forlin and Chambers, 2011). Teachers' attitudes towards inclusive education must be investigated in depth to determine how such attitudes affect their intentions

and actual practices in inclusive education, either directly or indirectly, through, for example, teachers' concerns related to inclusive education (Jury et al., 2023), self-efficacy (Miesera et al., 2019), or their expectations of students with disabilities (Moberg, 2003).

Researchers have found a relationship between perception of school support and the intention for inclusive education (Ahmmed, 2013; Hellmich et al., 2019). In this study, we found that perception of school support was not a predictor of inclusive educational intention. This result might be explained by the idea that teachers may not see how the support that might be provided would help with their practice in teaching students with disabilities, when the opportunity to implement inclusive education arrives. Indeed, the participants reported that they were unsure about the level of support provided when inclusive education was implemented. Thus, it is important to involve teachers in discussing what type of support they are willing to provide and in implementing inclusive education practices in the classroom, school administration, preparations of school personnel, and at institutional levels (Almalky & Alwahbi, 2023).

We expected teachers' perceptions of concern related to inclusive education to be associated with their intentions toward inclusive education. However, we were surprised to find that teachers' concerns were not predictors of their intentions to implement inclusive education in regular classrooms. One explanation is that teachers believe in their confidence to implement inclusive education in regular classrooms may reduce the impact of their concerns about inclusive education (Ahsan, et al., 2012; Miesera et al., 2019; Woodcock et al., 2012). Another explanation could be that the teachers in this study linked their potential concerns when implementing inclusive education with the availability of support and resources, which is only one aspect of their concerns. Studies have indicated that lack of resources was the greatest concern teachers reported, among other concerns such as workload, classroom management, academic standards, and acceptance (Alqahtani & Rababah, 2019; Warnes et al., 2022). It would be interesting to unpack different types of concerns and examine how they impact teachers' intentions to teach in an inclusive education classroom, and how they could impact their inclusive education practices, especially as concern has been found to correlate negatively with implementing inclusive education practices (Sharma & Sokal, 2016).

5. Limitations

The results of this study should be interpreted with caution. First, the sample in this study was small in size and voluntarily participated. We could not compare the sample to the population or collect the response rate. Thus, the results cannot be generalised to a larger population. Secondly, the study design was based on cross-sectional surveys. Therefore, causality could not be established between the variables. Third, this study utilised self-report measures, and the results may have been subject to social desirability bias and response uncertainty.

5.1. Implication

This study aimed to explore Saudi teachers' intentions towards inclusive education and the variables that could predict their intentions. In Saudi Arabia, the current practice of inclusive education focuses on certain types of disabilities, while other disabilities are not included. Such a situation may affect teachers' intentions. Thus, future research should examine whether the type of disability affects teachers' intentions regarding inclusive education. Furthermore, support and concern were correlated with intention but lost their association after including other variables, such as self-efficacy, in the model. We encourage further research to examine the interrelationships among these variables and how they may influence each other and ultimately influence teachers' intentions. Researchers should examine how teachers' attitudes differ across different types of disabilities in regular classrooms. Furthermore, it is important to examine how teachers'

attitudes (towards inclusive education and towards disability) might influence other aspects of inclusive education, such as advocacy for inclusive education and the perception of barriers and concerns.

6. Conclusion

In inclusive education, all students, including those with disabilities, are provided with equal opportunities and respect. However, teachers' intentions to implement inclusive education play a critical role. Several factors influence teachers' intentions, including their attitude, self-efficacy, perception of support, and concerns about inclusive education. Professional development training and preparation of teachers should consider these factors to make them positive influencers and increase teachers' knowledge of inclusive education practices, such as teaching strategies, evaluation, and differentiation of curricular content so that they can successfully overcome barriers and implement inclusive education in their classrooms.

CRediT authorship contribution statement

Hussain A. Almalky: Conceptualization, Funding acquisition, Investigation, Methodology, Validation, Resources, Writing – original draft, Writing – review & editing. **Abdalmajeed H. Alrabiah:** Data curation, Formal analysis, Investigation, Methodology, Validation, Resources, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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